Reg. No. | | | | | | | | | | | [18MEOE04]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

AUTOMOBILE ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	List down the main components of automobile chassis and explain its layout.	**07**
	b.	Briefly explain the fuel injection process for SI Engines.	**07**
2.	**a.**	Briefly, explain the working principal of torque conversion and fluid flywheel.	**06**
	b.	Explain the suspension system and its types.	**08**
3.	**a.**	Define wheel assembly and explain in detail the types of wheels.	**10**
	b.	Distinguish between wheel and tyre.	**04**
		OR	
4.	**a.**	What are the types of electric cars? Briefly explain the inner parts of an electrically driven vehicle.	**10**
	b.	Briefly explain the CNG operated vehicles.	**04**
5.	**a.**	Write short note on ABS and traction control.	**08**
	b.	Briefly explain the mechanics of moving vehicles.	**06**
		OR	
6.	**a.**	Briefly explain the mechanics involved in the motion of the vehicle on **i.** Level curved path **ii.** Banked circular track	**14**
7.	**a.**	What are the different pollutants from an automobile and explain formation process of any 3 pollutants.	**08**
	b.	Briefly explain the smoke due to combustion of an automobile.	**06**
		OR	
8.	**a.**	Explain various causes for noise pollution and measures to control noise pollution.	**08**
	b.	Discuss any two types of emission control systems automobiles.	**06**